



BANNARI AMMAN INSTITUTE OF TECHNOLOGY

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22PH202 – ELECTROMAGNETISM AND MODERN PHYSICS

Unit-V: ENERGY BANDS IN SOLIDS_LM2

Periodic Table of the Elements

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hydrogen
alkali metals
alkali earth metals
transition metals

post-transition metals
nonmetals
noble gases
halogens
metalloids

1 H																	2 He	
3 Li	4 Be																	10 Ne
11 Na	12 Mg																	18 Ar
19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr	
37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe	
55 Cs	56 Ba	57-71	72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn	
87 Fr	88 Ra	89-103	104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds	111 Rg	112 Cn	113 Uut	114 Fl	115 Uup	116 Lv	117 Uus	118 Uuo	

lanthanoids

57 La	58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb	71 Lu
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actinoids

89 Ac	90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No	103 Lr
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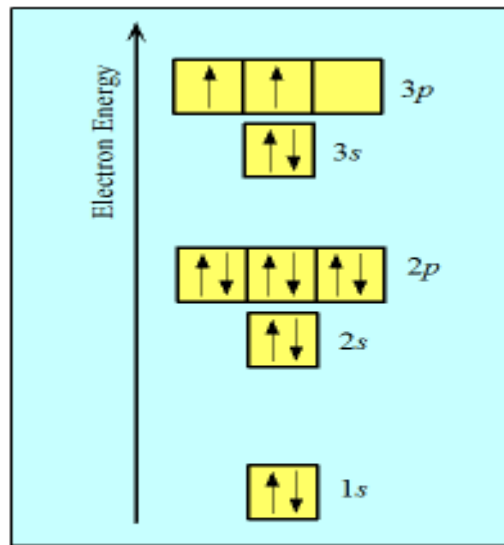
Elemental Semiconductors

IB	IIB	IIIA	IVA	VA	VIA	VIIA	VIIIA
29	30	31	32	33	34	35	36
47	48	49	50	51	52	53	54
79	80	81	82	83	84	85	86
111	112	113	114	115	116	117	118
Cu	Zn	Ga	Ge	As	Se	Br	Kr
Ag	Cd	In	Sn	Sb	Te	I	Xe
Au	Hg	Tl	Pb	Bi	Po	At	Rn

Element	Band Gap (eV)
Germanium (Ge)	0.67
Silicon (Si)	1.12
Carbon (C)	5.47

Z	Element	No. of electrons/shell
6	Carbon	2, 4
14	Silicon	2, 8, 4
32	Germanium	2, 8, 18, 4





14

Silicon

Si



Atomic mass: 28.085

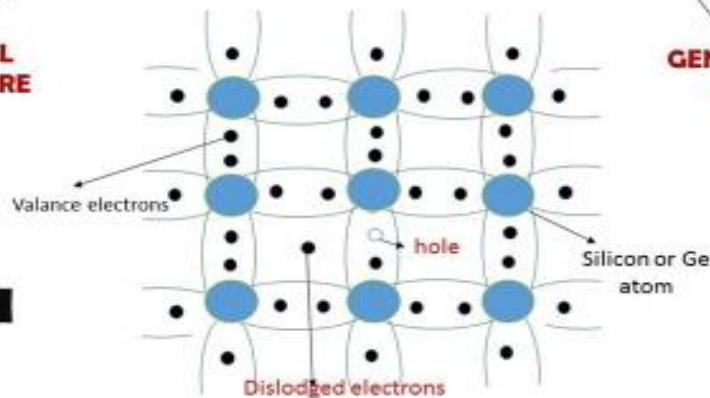
Electron configuration: 2, 8, 4

The electronic structure of Si.

INTRINSIC SEMICONDUCTOR

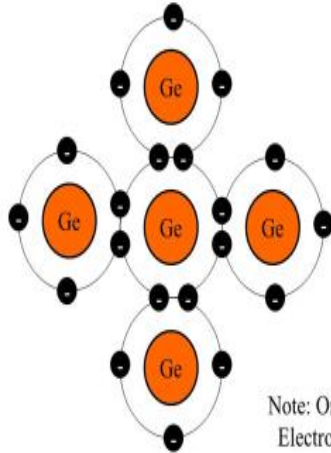
**CRYSTAL
STRUCTURE**

**GENERATION OF HOLES
AND ELECTRONS**



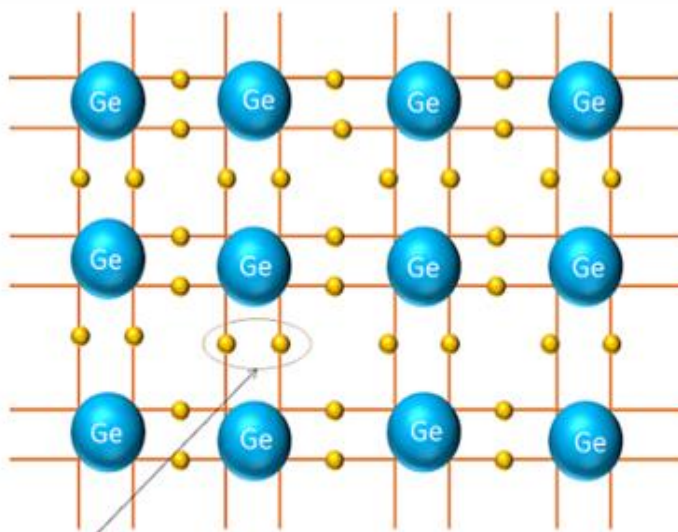
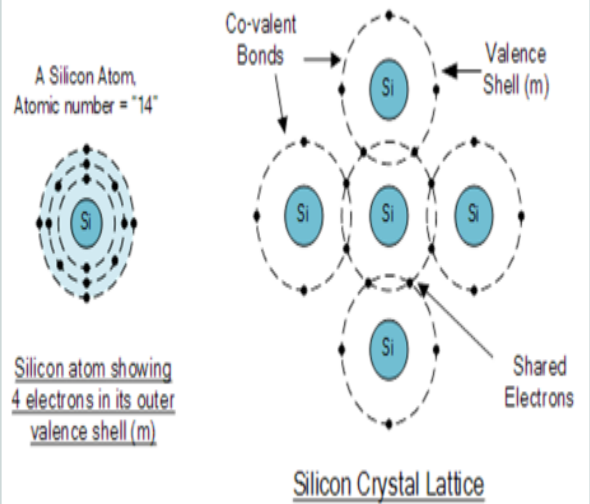
**IN
HINDI**

Germanium Covalent Bond



Note: Only Valence Electrons shown.

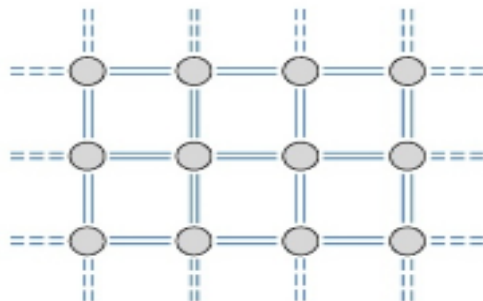
Bonding in Silicon



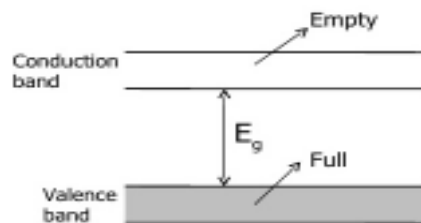
Sharing of electrons

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$T=0K$



(a) Covalent bonding in Si



(b) line representation of energy band diagram

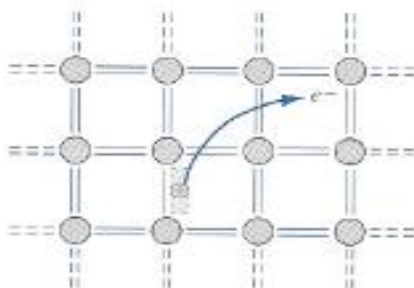
At $T = 0 K$, all the valence electrons are covalently bonded

At $T = 0 K$, Silicon act as a perfect insulator (not sufficient thermal energy)

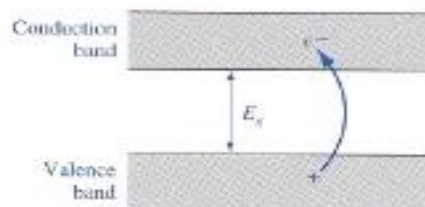
To provide conduction electrons, covalent bonds should be broken

Energy needed : **1.1 eV for Si** and **0.7 eV for Ge**

$T>0K$



(a) Two dimensional representation of breaking of a covalent bond.



(b)

At $T>0K$, some covalent bonds break giving rise to positively charged empty states and electrons

At room temperature, thermal energy is sufficient to break covalent bonds

When a covalent bond is broken, an electron goes to the conduction band leaving behind an empty space in the valence band (hole)

Equal number of electrons (conduction band) & holes (valence band)

Extrinsic Semiconductors

1. N – type Semiconductors
2. P – type Semiconductors



						VIIIA
						2 He 4.003
		IIIA	IVA	VA	VIA	VIIA
		5	6	7	8	9
		B	C	N	O	F
		10.811	12.011	14.007	15.999	18.998
						10 Ne 20.183
		13	14	15	16	17
		Al	Si	P	S	Cl
		26.982	28.086	30.974	32.064	35.453
						18 Ar 39.948
IB	IIB					
29	30	31	32	33	34	35
Cu	Zn	Ga	Ge	As	Se	Br
63.54	65.37	69.72	72.59	74.922	78.96	79.909
47	48	49	50	51	52	53
Ag	Cd	In	Sn	Sb	Te	I
107.870	112.40	114.82	118.69	121.75	127.60	126.904
79	80	81	82	83	84	85
Au	Hg	Tl	Pb	Bi	Po	At
196.967	200.59	204.37	207.19	208.980	(210)	(210)
						86 Rn (222)

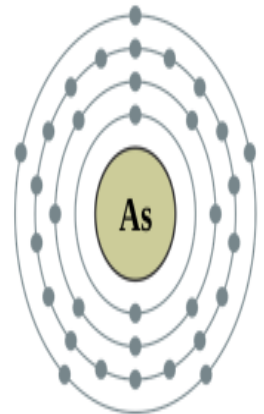
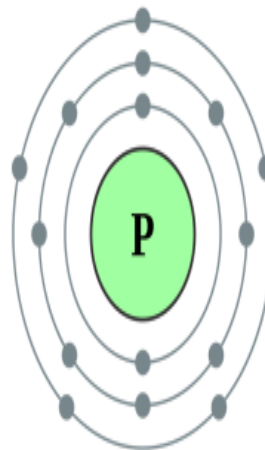
N-type Semiconductor

Pentavalent Element

Element	Atomic Number
Phosphorous (P)	15
Arsenic (As)	33
Antimony (Sb)	51

Donor atoms

	13 IIIA 3A	14 IVA 4A	15 VA 5A
5	10.811	12.011	14.007
B	Boron	C	Carbon
[He]2s ² 2p ¹		[He]2s ² 2p ²	
13	26.982	28.086	30.974
Al	Aluminum	Si	Silicon
[Ne]3s ² 3p ¹		[Ne]3s ² 3p ²	
31	68.723	72.631	74.922
Ga	Gallium	Ge	Germanium
[Ar]3d ¹⁰ 4s ² 4p ¹		[Ar]3d ¹⁰ 4s ² 4p ²	
49	114.818	118.711	121.760
In	Indium	Sn	Tin
[Kr]4d ¹⁰ 5s ² 5p ¹		[Kr]4d ¹⁰ 5s ² 5p ²	
		51	121.760
		Sb	Antimony
		[Kr]4d ¹⁰ 5s ² 5p ³	



N-type Semiconductor

An Antimony Atom,
Atomic number = "51"



Antimony atom showing
5 electrons in its outer
valence shell (o)

